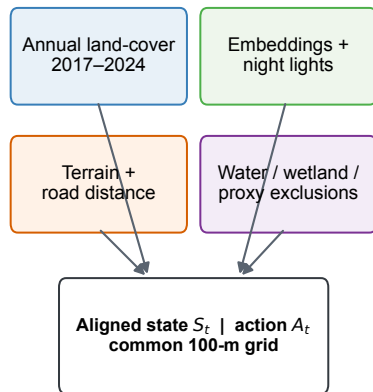
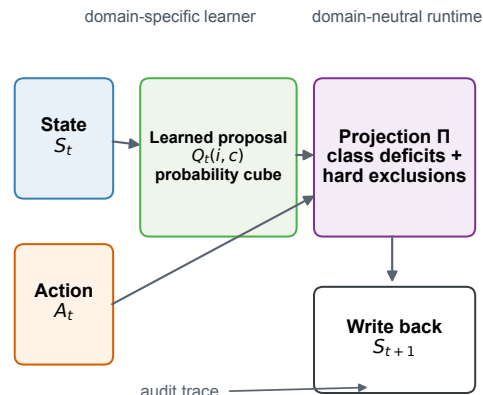


A common state–action–constraint makes spatial allocation auditable

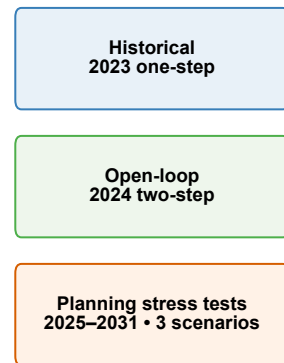
A Common public-data benchmark



B Geospatial Kernel execution loop



C Evaluation tracks



same evaluator

What is shared
State schema
Action semantics
Constraint audit
Output provenance

What is model-specific
Proposal learner
Feature representation
Transition dynamics
Allocation score

Interpretation boundary
Public land cover $\neq$ legal land use
Proxy masks $\neq$ statutory red lines
Scenario maps $\neq$ official forecasts